The Invoice App
Database Architecture & Schema

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Overview

This document outlines the Relational Database structure for the Invoice App. Instead of utilizing multiple floating spreadsheet files, the system uses a single SQLite database (MasterDB.db) containing four interconnected tables. This guarantees data integrity, prevents duplicate entries, and allows for complex financial reporting (like tracking total profits while ignoring cancelled orders).

1 Products (The Inventory / Menu)

Functionality: This table acts as the central menu. It holds all available items for sale, their respective retail and wholesale pricing, and an optional stock counter. As seen in the vacuum cleaner script, it seamlessly absorbs data directly from the original CSV files while maintaining legacy product IDs.

Field Name	Data Type	Description
Product_ID	INTEGER (PK)	Unique identifier for each item. Explicitly matches the original Excel/CSV prod_ID.
item_name	TEXT	The primary name or description of the product.
description	TEXT	Additional details about the item (currently kept blank as it is merged with item_name).
Retail_Price	REAL	The standard internal/retail price of the item.
Wholesale_Price	REAL	The discounted or bulk price of the item.
stock_quantity	INTEGER	Optional inventory tracking. Defaults to 0 so the app doesn't crash if stock tracking is disabled in the YAML config.
Cost	REAL DEFAULT 0.0	The baseline cost of the item.

2 Customers (The Client Book)

Functionality: Keeps track of everyone who buys from you. This prevents having to type out contact information manually for every invoice and remembers whether a specific customer should automatically receive wholesale or retail pricing.

Field Name	Data Type	Description
customer_id	INTEGER (PK)	Unique, auto-incrementing ID for the customer.
Name	TEXT NOT NULL	The full name of the customer or business.
Phone_Number	TEXT	Contact information for the client.
Default_Tier	TEXT	Automatically flags the customer as 'retail' or 'wholesale' to quickly apply the correct pricing matrix.

3 Orders (The Invoice Headers)

Functionality: This is the financial core. It represents the actual "Bill" itself. It stores the grand totals, the date of purchase, and the overall profit. Crucially, it features a status column that allows for reversibility—meaning an invoice can be "cancelled" without deleting the record entirely, keeping invoice numbers perfectly sequential.

Field Name	Data Type	Description
Invoice_Number	INTEGER (PK) AUTOINCREMENT	The official Invoice Number.
Customer_ID	INTEGER NOT NULL (FK)	Links directly to the Customers table.
Date	TEXT NOT NULL	Timestamp of when the invoice was generated.
Subtotal	REAL NOT NULL	Total amount before any cart-level discounts.
Discount	REAL NOT NULL	Overall discount applied to the entire invoice.
Total	REAL NOT NULL	The final amount the customer actually owes.
Profit	REAL NOT NULL	Calculated field representing the profit margin for the specific order.
Status	TEXT NOT NULL	Defaults to 'active'. Can be changed to 'cancelled' to reverse the order without deleting data.

4 OrderDetails

Functionality: Since a single order can contain 50 different items, this table breaks down the invoice line by line. It links the master Orders table to the Products table, recording exactly what was bought, how many, and the exact price they were charged at that moment in time (protecting historical invoices if you raise your prices next year).

Field Name	Data Type	Description
Invoice_Number	INTEGER NOT NULL (FK)	Links this specific item to its parent Invoice/Order.
Item_ID	INTEGER NOT NULL (FK)	Links back to the Products table to identify what was sold.
Quantity	INTEGER NOT NULL	The number of units purchased for this specific item.
Price_Sold	REAL NOT NULL	The actual price charged at the time of sale (locks in the price to prevent historical changes).